Christian Joudon

SOFTWARE ENGINEER · TECHNICAL WRITER · FULL-STACK DEVELOPMENT · AI & AUTOMATION

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Portfolio · Technical Writing & Software christianjoudon.github.io

SUMMARY

Software engineer and technical writer with experience in full-stack applications, automation, and data systems. I work across product design, databases, backend services, and interfaces, and develop documentation that makes complex systems easier to understand and use.

EDUCATION

University of Hawaiʻi at Mānoa

2020 - 2024

B.S., Computer Science · GPA 3.79 · Oʻahu

Coursework: Data Structures, Machine Learning, Web Development, Database Systems.

EXPERIENCE

Accounting Operations & Systems Development

2025 - Present

Hawaii Life · Kīlauea, Hawaiʻi

- Designed and implemented AI-assisted email-to-Escapia workflows for reservation entry and updates, combining JavaScript browser automation, exact property matching, financial reconciliation, saved-state verification, and human review for financial submissions.
- Built channel-aware email parsers in Google Sheets for Airbnb, Booking.com, Marriott/HVMI, and direct bookings. Extracted reservation details, taxes, commissions, and payouts; validated formats and financial totals across 66 native spreadsheet QA cases.
- Developed an Apps Script insurance workbench and responsive document-upload portal linking property records, certificate collection, Drive storage, and Sheets synchronization, with unit validation, upload checks, duplicate detection, and personalized owner correspondence.
- Expanded refund report ingestion, receipt matching, and Gmail draft preparation. Hardened Apps Script batch storage with compressed, chunked payloads and SHA-256 integrity checks while preserving existing reports and completed correspondence.
- Delivered a recurring Python ACH document processor using qpdf, Ghostscript, and Tesseract OCR. Assembled multipage receipts, matched vendors, reconciled payment totals, and published validated packets to an Apps Script queue for Gmail drafts; verified with 34 processor tests.
- Built an AI-assisted invoice dashboard that retrieves email and brings extracted fields, accounting memos, and source documents into one review table for staff entry into Escapia; standardized document naming and filing.
- Authored nine illustrated operational handbooks totaling 300+ pages, with annotated screenshots, linked references, decision paths, and exception handling for reservations, accounting, refunds, and insurance.

Software Engineer

2025 - Present

HīTech Labs · Kauaʻi, Hawaiʻi

- Develop independent software products for rental operators, service businesses, and local communities, including HAVEN, Chime, CheckInn, and MatchBook. Work across product design, data modeling, backend services, interactive interfaces, and product documentation.

TECHNICAL AND EDITORIAL SKILLS

Languages and application development: Python, TypeScript, JavaScript, SQL, HTML/CSS, React, Next.js, Node.js, Express, FastAPI, NestJS

Data and integrations: PostgreSQL, MongoDB, SQLAlchemy, REST APIs, webhooks, Docker, Google Apps Script, Gmail/Drive APIs, Google Sheets, Playwright, OCR

AI and documentation: LLM extraction, Pydantic, FAISS, PySpark, Pandas; technical explanation, tutorials, SOPs, information design, structural editing

SELECTED SOFTWARE AND DATA PROJECTS

HAVEN · Accounting for rental businesses

Python · FastAPI · SQLAlchemy · PostgreSQL · React · TypeScript

Built an accounting platform prototype connecting bills, payments, bank reconciliation, and owner distributions through a shared ledger. Document extraction and batch review link source files to financial records; validation and approval controls support traceable accounting workflows.

Case study christianjoudon.github.io/work/haven.html

Chime · Booking and business management

TypeScript · React · Node.js / Express · PostgreSQL · Stripe Connect

Developed a customer booking flow and business workspace around one backend. Server-side availability checks precede confirmation; Stripe Connect supports deposits and refunds. Appointment records connect customer details, schedule changes, payment history, and email delivery.

Website bookwithchime.com · Case study christianjoudon.github.io/work/chime.html

CheckInn · Reservations and property operations

TypeScript · Next.js / React · NestJS / Fastify · PostgreSQL

Built a rental operations platform connecting available nights, guest records, reservations, and property readiness. A shared booking engine manages holds, quotes, and confirmed stays while retaining accepted terms. Arrival and housekeeping views organize the work required for each stay.

Case study christianjoudon.github.io/work/checkinn.html

MatchBook · Local events and community discovery

Meteor · React · MongoDB · TypeScript · Leaflet · FullCalendar

Built a Kauaʻi discovery app with feeds, swipe cards, maps, and calendars for local events and community groups. Interest and membership signals rank listings; ingestion and review prepare them for publication, and saved activities form a personal agenda.

Case study christianjoudon.github.io/work/matchbook.html

Medical records · AI-assisted data analysis

Python · LLM extraction · Pydantic · FAISS · PySpark

Built a pipeline that turns free-text medical notes into structured records for studying COVID-19 patient patterns. Pydantic validates extracted fields, FAISS supports medication matching, and PySpark aggregates patterns in related conditions, medications, and demographics.

Portfolio christianjoudon.github.io

SELECTED TECHNICAL WRITING

Product engineering case studies. The HAVEN and Chime case studies explain architecture, data flows, financial controls, and interface decisions through complete user workflows. Samples are linked above.

Operational guides and walkthroughs. Translate complex procedures into step-by-step instructions, screenshot-based references, and review criteria for Hawaii Life staff. Cover routine work and exceptions; samples available on request.

A Near Tie, Decades in the Making. Interactive research essay tracing the changing balance of renewable and coal-fired electricity generation.

Essay christianjoudon.github.io/essays/a-near-tie.html

ADDITIONAL EXPERIENCE

Creative Writer · Freelance · 2019 - Present
Real Estate Marketing Copywriter · Freelance · 2019 - 2021
Project Manager · Bacchus LLC · 2018 - 2020
Food & Hospitality · Multiple establishments · 2010 - 2018

LANGUAGES

English (Native) · French (Proficient) · German (Intermediate)